

ROBOTIC ARM CONTROLLED BY HAND GESTURES

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Abstract—A Robotic Arm controlled by a Robotic Glove using Flex Sensor is a promising development in the field of robotics that offers a natural and intuitive way for humans to control complex machinery. This system consists of a glove worn by the human operator and a robotic arm that mimics the movement of the operator's hand. The glove is equipped with flex sensors that detect the movement of the fingers and hand and transmit this data to the robotic arm. The robotic arm then replicates the motion of the operator's hand in real-time, allowing for precise control of the machinery. One of the key advantages of this system is its ease of use. The flex sensors on the glove provide a reliable and cost-effective way to capture the motion of the operator's hand, making it an attractive option for a wide range of industries. Additionally, the system is designed to be intuitive, allowing operators to control complex machinery with minimal training. This is particularly important in industries such as manufacturing and medical procedures, where operators must be able to perform intricate tasks with high precision. Another advantage of the Robotic Arm controlled by a Robotic Glove using Flex Sensor is its potential for use in remote control of machinery. The system can be used to control machinery from a distance, allowing operators to perform tasks in hazardous or hard-to-reach locations. This could be particularly useful in industries such as mining, oil and gas, and construction, where workers often operate in dangerous environments.

Index Terms—Arduino uno, Arduino Nano, Flex Sensor, Accelerometer, Bluetooth HC-05 module.

I. INTRODUCTION

The rapid evolution of robotics and embedded systems has accelerated the integration of intelligent automation into industrial, medical, and research domains. Robotic manipulators are increasingly employed in applications requiring precision, repeatability, and operational safety. However, conventional control techniques such as joysticks, switches, and pre-programmed interfaces often lack intuitive interaction and demand significant operator training. As task complexity grows, these traditional control mechanisms may reduce efficiency and limit natural human-machine collaboration. Consequently, there is a growing need for alternative control strategies that

provide seamless, real-time interaction between humans and robotic systems.

Gesture-based robotic control has emerged as an effective approach to address these challenges by enabling machines to interpret natural human movements. In such systems, wearable sensing devices incorporating flex sensors and motion sensors detect finger bending and wrist orientation, converting physical gestures into corresponding electrical signals. These signals are processed using microcontroller platforms and transmitted wirelessly to a robotic arm unit, where actuators reproduce the intended motion. Since signal acquisition and processing occur at high speeds, the robotic arm can respond with minimal delay, allowing synchronized and smooth operation. This method enhances user convenience and eliminates dependence on complex mechanical interfaces.

One of the major advantages of gesture-controlled robotic systems is their intuitive operation, which significantly reduces learning time and improves productivity. Wireless communication further enables remote control capabilities, making the system suitable for environments that are hazardous, inaccessible, or sensitive to direct human intervention. Additionally, the use of low-cost microcontroller-based architectures ensures economic feasibility and ease of implementation without compromising functional performance.

Despite these benefits, system performance is influenced by factors such as sensor calibration accuracy, signal noise, communication latency, and precise mapping of gestures to actuator movements. Careful system design and testing are therefore essential to ensure reliable operation and consistent performance. In this work, a hand gesture-controlled robotic arm is designed and implemented using embedded sensing elements, wireless communication modules, and servo-driven joints. Experimental validation is carried out to evaluate responsiveness, motion accuracy, and overall system reliability, demonstrating the practicality of gesture-based control for modern robotic applications.

II. LITERATURE REVIEW

Gesture-based robotic manipulation has been widely explored as an alternative to conventional joystick and teach-pendant control systems. Vision-based gesture-driven robotic arm frameworks were introduced in [1] and [2], demonstrating the feasibility of natural hand gesture interfaces for industrial and assembly-line applications. These studies emphasized improved human–robot interaction and operational convenience; however, their validation was largely confined to controlled environments with limited analysis of robustness, scalability, and computational efficiency under dynamic conditions. Such limitations highlight the need for optimized real-time gesture recognition systems with enhanced adaptability.

Communication-enhanced robotic architectures have also been investigated to improve coordination and system-level intelligence. Digital twin-based multimodal human–robot interaction was explored in [3], while IoT-enabled robotic automation frameworks were presented in [4]. Although these approaches enhanced monitoring, synchronization, and remote accessibility, increased network dependency and computational overhead introduced latency and reliability concerns, particularly in time-sensitive applications. These challenges emphasize the importance of designing lightweight and low-latency control mechanisms suitable for embedded robotic systems.

Low-cost and microcontroller-based implementations of gesture-controlled robotic arms were developed in [5] and [6], utilizing Arduino platforms and mobile-integrated systems. These approaches demonstrated affordability and ease of implementation; however, constraints related to payload capacity, positioning precision, and sensor accuracy were observed. The trade-off between cost-effectiveness and performance therefore remains a critical design consideration. Advanced intelligent control strategies incorporating artificial potential fields and deep reinforcement learning were proposed in [7], while structural optimization and industrial robotic arm enhancements were discussed in [8]. Although these methods improved adaptability and planning efficiency, they required higher computational resources and increased system complexity, limiting their suitability for lightweight gesture-based systems.

Safety-oriented articulated robotic arms for hazardous material handling were presented in [9], emphasizing precision and secure operation in controlled environments, though implementation complexity and cost remained significant challenges. Additionally, glove-based robotic arm control systems were explored in [10], demonstrating practical feasibility with wearable sensing. Despite substantial progress in gesture recognition, communication frameworks, intelligent control, and mechanical optimization, existing systems continue to face limitations such as environmental sensitivity, communication delay, hardware constraints, and computational overhead. Therefore, there remains a clear need for a cost-effective and efficient gesture-controlled robotic arm system capable of reliable performance under practical operating conditions.

III. HARDWARE IMPLEMENTATION

The hardware implementation of the proposed gesture-controlled robotic arm system consists of two major units: a wearable transmitter module and a robotic arm receiver module. The system enables real-time wireless control of a multi-degree-of-freedom robotic arm using natural hand gestures.

A. System Architecture

The overall system architecture consists of four primary blocks: sensing, processing, wireless communication, and actuation, as shown in Fig. 1. The sensing block includes flex sensors and a three-axis accelerometer mounted on a wearable glove. The processing block converts analog sensor readings into calibrated digital values. Wireless communication is achieved using HC-05 Bluetooth modules configured in master–slave mode. The actuation block consists of multiple servo motors forming the joints of the robotic arm.

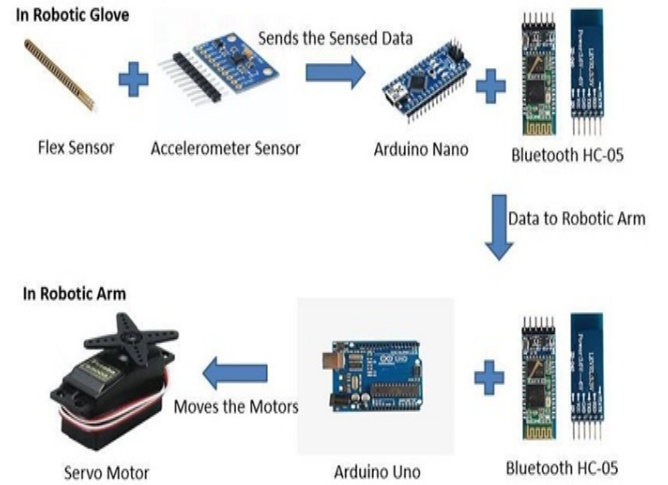


Fig. 1. Overall System Architecture of the Proposed Robotic Arm

B. Transmitter Module

The transmitter module serves as the master unit in the gesture-controlled robotic arm system and is implemented as a wearable robotic glove that captures human hand motion for real-time control. The glove integrates multiple flex sensors and a three-axis accelerometer to detect finger bending and wrist orientation, where resistance changes in the flex sensors correspond to the degree of finger motion and inertial measurements hand rotation and tilt. As shown in Fig. 2, the transmitter hardware consists of the Arduino Nano, sensor interfacing circuitry, and the HC-05 Bluetooth module mounted on a compact development board for real-time gesture acquisition. Analog sensor outputs are acquired through voltage divider circuits and fed into the microcontroller, which continuously samples and calibrates the readings to derive meaningful gesture parameters.

These parameters are mapped to predefined control values representing specific arm movements, such as base rotation, gripper actuation, and wrist movement. The processed data is transmitted wirelessly to the robotic arm via an HC-05 Bluetooth module configured in master mode, establishing a low-latency communication link for coordinated gesture recognition. A rechargeable battery powers the unit to ensure portability, and careful circuit design is employed to minimize electrical noise and improve signal fidelity.

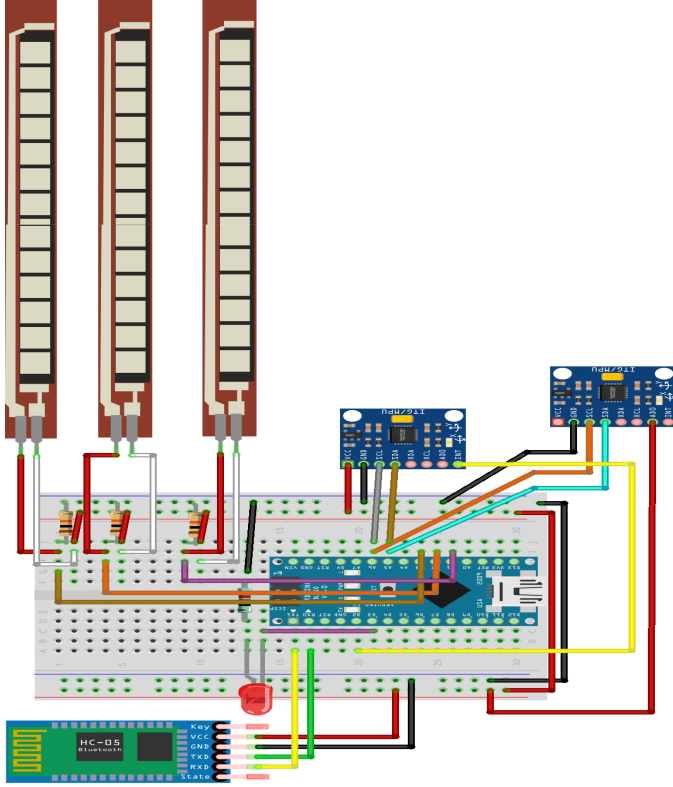


Fig. 2. Transmitter Module Block Diagram

C. Receiver Module

The receiver module operates as the slave unit and is responsible for interpreting gesture data transmitted from the master and converting it into mechanical actuation. The unit is based on an Arduino Uno microcontroller interfaced with an HC-05 Bluetooth module configured in slave mode, which continuously listens for incoming serial data. As illustrated in Fig. 3, the receiver hardware setup comprises the Arduino board, motor driver circuitry, regulated power supply, and multiple servo motors assembled for joint actuation of the robotic arm.

Upon reception, the decoded values are mapped to corresponding servo and stepper motor commands, where each joint of the robotic arm is controlled according to the gesture mapping table defined in the system architecture. PWM signals generated by the Arduino and, if used, an external servo driver are applied to the respective actuators to achieve coordinated

motion across multiple degrees of freedom, such as shoulder articulation, elbow bending, and gripper manipulation.

Dedicated power regulation ensures that high-current servos receive stable supply without affecting microcontroller logic levels, while isolation between logic and motor power minimizes interference. This master-slave communication framework enables reliable, real-time command execution, facilitating seamless translation of hand gestures into precise robotic movements.

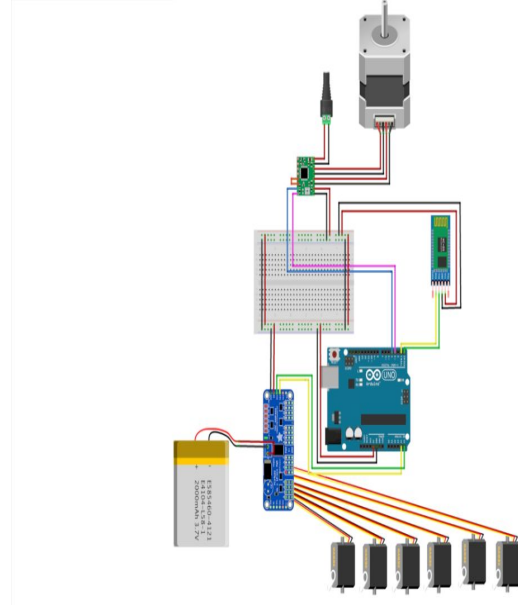


Fig. 3. Receiver Module and Robotic Arm Assembly

D. Experimental Analysis

The experimental validation of the proposed hand gesture-controlled robotic arm was performed under controlled laboratory conditions to ensure systematic testing and accurate performance assessment. The hardware configuration, including the transmitter and receiver modules, was carefully examined during experimentation. Arduino IDE Serial Monitor outputs were recorded to verify communication stability, gesture interpretation accuracy, and overall system responsiveness during real-time operation.

(a) Hardware Setup and Command Mapping:

The complete system consists of a wearable transmitter glove based on Arduino Nano and a receiver module built on Arduino Uno. The glove integrates multiple flex sensors and dual MPU6050 accelerometers to accurately capture finger bending and wrist orientation in real time. The receiver unit controls multiple servo motors through a PCA9685 driver and a stepper motor for precise base rotation. Independent regulated power supplies were implemented to ensure voltage stability, minimize electrical noise, and prevent servo-induced fluctuations during continuous operation. The receiver module and complete robotic arm assembly used during experimentation are

shown in Fig. 4. The gesture-to-command mapping used for wireless transmission is summarized in Table I.

TABLE I
GESTURE CONDITION AND CHARACTER MAPPING

Gesture Condition	Transmitted Character
Wrist Left	L
Wrist Right	R
Wrist Up	G
Wrist Down	U
Shoulder Control	C / c
Pinkie Bend	P / p
Thumb Bend	T / t
Finger Bend	F / f

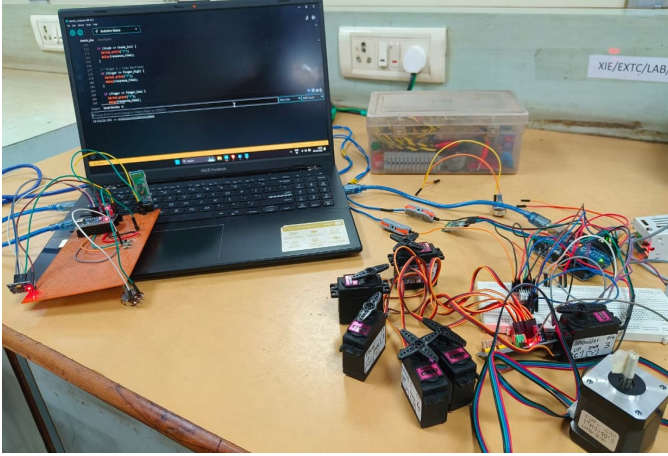


Fig. 4. Receiver Module and Robotic Arm Assembly

(b) Serial Monitor Observation:

The Arduino IDE Serial Monitor output confirms stable wireless communication and correct gesture interpretation, as shown in Fig. 5. Repeated characters observed during experimentation indicate continuous gesture detection and reliable data transmission between the transmitter and receiver modules.

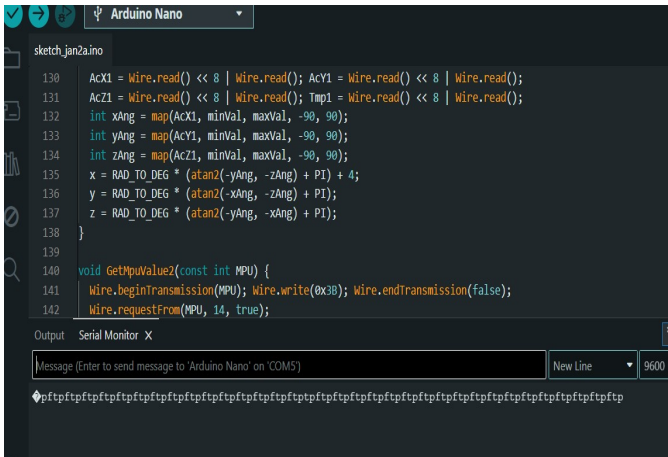


Fig. 5. Command Response on Serial Monitor

(c) Motion Response and Performance:

Upon receiving command characters, the receiver executes predefined motion routines. Servo motors operate within constrained angular limits to prevent mechanical over-rotation, while the stepper motor generates controlled base movement. The practical motion behavior of the robotic arm corresponds to the hardware configuration shown in Fig. 4. Experimental results demonstrate smooth motion, minimal response delay, and consistent gesture-to-action mapping.

Multiple experimental trials confirmed reliable operation, stable Bluetooth connectivity, and repeatable joint positioning. The system achieved real-time gesture recognition with low latency, validating the effectiveness of the master-slave embedded architecture for practical robotic control applications.

IV. RESULTS AND ANALYSIS

The proposed hand gesture-controlled robotic arm was tested under multiple operating conditions to evaluate its performance. The system successfully recognized different hand gestures and translated them into corresponding robotic arm movements in real time. Basic operations such as gripper opening and closing, elbow bending and extension, wrist rotation, and base movement were executed consistently across repeated trials as shown in Fig. 6.

The gesture recognition accuracy was satisfactory, and the response time between gesture input and robotic arm movement was minimal, enabling smooth operation. Stable and precise motion was observed when a reliable Bluetooth connection was maintained. Minor delays occurred under wireless interference or weak signal conditions; however, these did not significantly affect overall system performance. The robotic arm resumed stable operation after Bluetooth reconnection without abnormal movements.

The results confirm that the proposed system provides an effective and intuitive solution for controlling robotic manipulators, making it suitable for industrial automation such as Pick and Place and remote handling applications.



Fig. 6. Receiver Module and Robotic Arm Assembly

V. ADVANTAGES AND LIMITATIONS

The proposed hand gesture-controlled robotic arm offers several advantages. The system provides a natural and intuitive human-machine interface, enabling operators to control complex robotic movements using simple hand gestures. This eliminates the need for conventional control devices such as joysticks or keypads and significantly reduces the learning curve. Real-time feedback obtained from wearable flex sensors and accelerometers allows precise control, making the system suitable for delicate and repetitive operations. Furthermore, wireless communication between the gesture input unit and the robotic arm enhances operational flexibility and improves operator safety by allowing control from a distance, particularly in hazardous or inaccessible environments.

Despite these advantages, the system has certain limitations. Gesture recognition accuracy is highly dependent on proper sensor calibration and consistent hand positioning. External factors such as sensor drift, wireless interference, and environmental noise can affect system performance. Additionally, the Bluetooth communication module offers a limited operating range, restricting long-distance applications. The payload capacity of the robotic arm is constrained by the torque capabilities of the servo motors, making the system unsuitable for heavy-duty industrial tasks. These limitations indicate the need for improved sensing accuracy, advanced communication technologies, and enhanced mechanical design in future developments.

VI. CONCLUSION AND FUTURE SCOPE

This paper presented the design and implementation of a hand gesture-controlled robotic arm using wearable sensing and embedded control techniques. The system enables natural and real-time interaction between the human operator and the robotic manipulator through the integration of flex sensors, motion sensing elements, microcontroller-based processing, and wireless communication. Experimental results demonstrated accurate gesture interpretation with minimal response delay, ensuring smooth and reliable robotic operation. The proposed solution provides a cost-effective and user-friendly alternative to conventional control methods, making it suitable for applications in industrial automation, remote handling, and assistive robotics.

Despite the successful implementation, the system offers significant opportunities for further enhancement. Integration of artificial intelligence and machine learning algorithms could enable adaptive gesture recognition and task learning capabilities, allowing the robotic arm to perform more complex and autonomous operations. Improvements in sensing technology may increase accuracy and responsiveness, leading to more precise motion control. The incorporation of long-range wireless communication can extend operational distance for use in hazardous or inaccessible environments. Enhancing load capacity and enabling multitasking capabilities would further expand industrial applicability. Additionally, with appropriate refinement and precision control mechanisms, the system can be adapted for advanced medical applications.

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REFERENCES

- [1] S. Abhishek, Y. S. Jogi, U. K. Sahu, S. K. Dash, and U. K. Yadav, “Teach pendant at fingertips: Intuitive vision-based gesture-driven control of dexterous robotic arm,” *IEEE Transactions on Mechatronics (Preprint)*, pp. 1–8, June 2025.
- [2] G. Angelidis and L. Bampis, “Gesture-controlled robotic arm for small assembly lines,” *Machines*, pp. 1–10, February 2025.
- [3] G. A. Al and U. Martinez-Hernandez, “Digetac unit for multimodal communication in human-robot interaction,” *IEEE Sensors Letters*, vol. 8, no. 4, pp. 650–654, April 2024.
- [4] S. Surya, K. Srinivasan, S. Jayanth, and R. Venkatesan, “Tot-enabled gesture-controlled robotic arm for enhanced industrial automation,” *International Journal of Advanced Engineering and Research Development (IJAERD)*, vol. 24, no. 1, pp. 230–238, March 2024.
- [5] S. A. Khan and V. Mundhada, “Arduino-based gesture-controlled robot,” *International Journal of Advanced Research in Science, Communication and Technology (IJAR SCT)*, vol. 5, no. 4, pp. 100–106, February 2025.
- [6] D. B., R. S., S. G., S. A., and S. Krishnapur, “Literature survey on gesture-controlled robotic arm with mobile rover integration,” *International Advanced Research Journal in Science, Engineering and Technology (IARJSET)*, vol. 11, no. 12, pp. 50–56, December 2024.
- [7] J. P., S. Altnji, K. Thiyagarajan, J. S. Nanda, A. Biswas, and A. R. Chowdhury, “Human-aware robot collaborative task planning using artificial potential field and dqn reinforcement learning,” *IEEE Robotics and Automation Letters*, vol. 8, no. 9, pp. 980–987, September 2023.
- [8] H. Lohakare, P. Jadhav, D. Wanjari, T. Chainde, K. Rahangdale, and A. Ikhar, “Human controlled robotic arm,” *International Research Journal of Modernization in Engineering Technology and Science (IR-JMETS)*, vol. 6, no. 3, pp. 401–407, March 2024.
- [9] F. A. Wasay, H. Ghose, and M. A. Rahman, “Gamora: A gesture articulated meta operative robotic arm for hazardous material handling in containment-level environments,” *International Journal of Advanced Research in Engineering and Technology (IJARET)*, vol. 16, no. 6, pp. 120–127, June 2025.
- [10] P. Penchalaiah, P. Akhil, P. Ramesh, P. N. S. P. Reddy, and M. N. Babu, “Robotic arm controlled by robotic glove,” *International Journal of Research in Engineering, IT and Social Sciences (IJREISS)*, vol. 13, no. 5, pp. 77–83, May 2023.